

Garvit Sonawala

 GarvitSonawala |  garvit22@iisertvm.ac.in |  +918318599831 |  garvit-sonawala.github.io

RESEARCH INTERESTS

Theoretical and computational cosmology, with a focus on early-Universe physics, dark matter phenomenology, high-energy theory, and gravitational-wave cosmology. My interests lie in developing and analyzing theoretical models of cosmic and particle phenomena, using computational and machine-learning techniques as tools for simulation, inference, and data analysis, while keeping the primary emphasis on advancing the underlying physical theory.

EDUCATION

Indian Institute of Science Education and Research (IISER) Thiruvananthapuram 2022 – 2027
BS-MS (Integrated) in Physics CGPA: 7.8

Indian Institute of Technology Madras 2023 – Present
BS in Data Science CGPA: 8.21

Delhi Public School, Varanasi 2020 – 2021
CBSE Class XII Percentage: 92.4

WORK EXPERIENCE

Minor Project in Chemistry, Indian Institute of Science Education and Research (IISER) Jan 2026 – Present
Supervisor: Dr. R. S. Swathi
Applied Particle Swarm Optimization (PSO) for global optimization of chemical geometries. Investigating swarm size, topology, and convergence behavior for complex molecular potential energy surfaces.

Student Intern, Tata Institute of Fundamental Research (TIFR), Mumbai May 2025 – Present
Supervisor: Dr. Achamveedu Gopakumar
Developed gravitational-wave detection models for circular SMBHBs using 12-year PTA residuals (0PN–2PN). Implemented CNN, RNN, and Bayesian NNs for waveform classification and chirp-mass estimation at low SNR.

Student Intern, Indian Institute of Technology Guwahati (IITG) Jul 2024 – Sep 2024
Supervisor: Dr. Subhaditya Bhattacharya
Modeled the time evolution of dark-matter relic density via the Boltzmann equation. Explored astrophysical evidence to compare theoretical predictions with observational data.

PRESENTATIONS

Exploring Dark Matter Freeze-Out ANVESHA 2024, IISER Thiruvananthapuram
Poster presentation based on my internship work, illustrating dark matter’s cosmological significance, freeze-out behaviour, and observational evidence.

VOLUNTEERING & LEADERSHIP

Core Committee Member, PSIT Physics Society, IISER TVM Aug 2024 – Present
Managing sponsorships, event logistics, merchandise, and content creation for departmental and outreach activities.

TECHNICAL SKILLS

Programming: Python, Java, SQL

Libraries / Tools: NumPy, Pandas, Matplotlib, Jupyter, PyTorch, scikit-learn, Git, LaTeX

Machine Learning: CNN, RNN, Bayesian Neural Networks, supervised & unsupervised learning, regression, clustering

Database / Web: MySQL, PostgreSQL, Flask, SQLAlchemy

RELEVANT COURSEWORK

Physics: Classical Mechanics, Quantum Mechanics I II, Electrodynamics, Statistical Mechanics, Special Relativity, General Relativity, Nuclear Physics, Astrophysics, Quantum Field Theory (*ongoing*), Particle Physics (*ongoing*)

Mathematics: Linear Algebra, Multivariable Calculus, Probability & Statistics

Programming / Data Science: Machine Learning, Data Analysis & Visualization